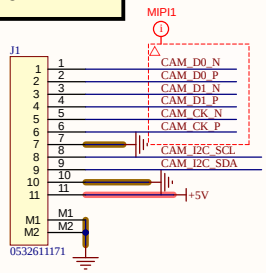
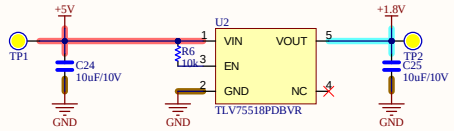
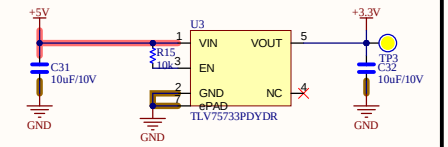
to OBC



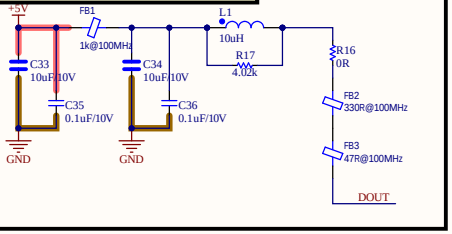
Connector



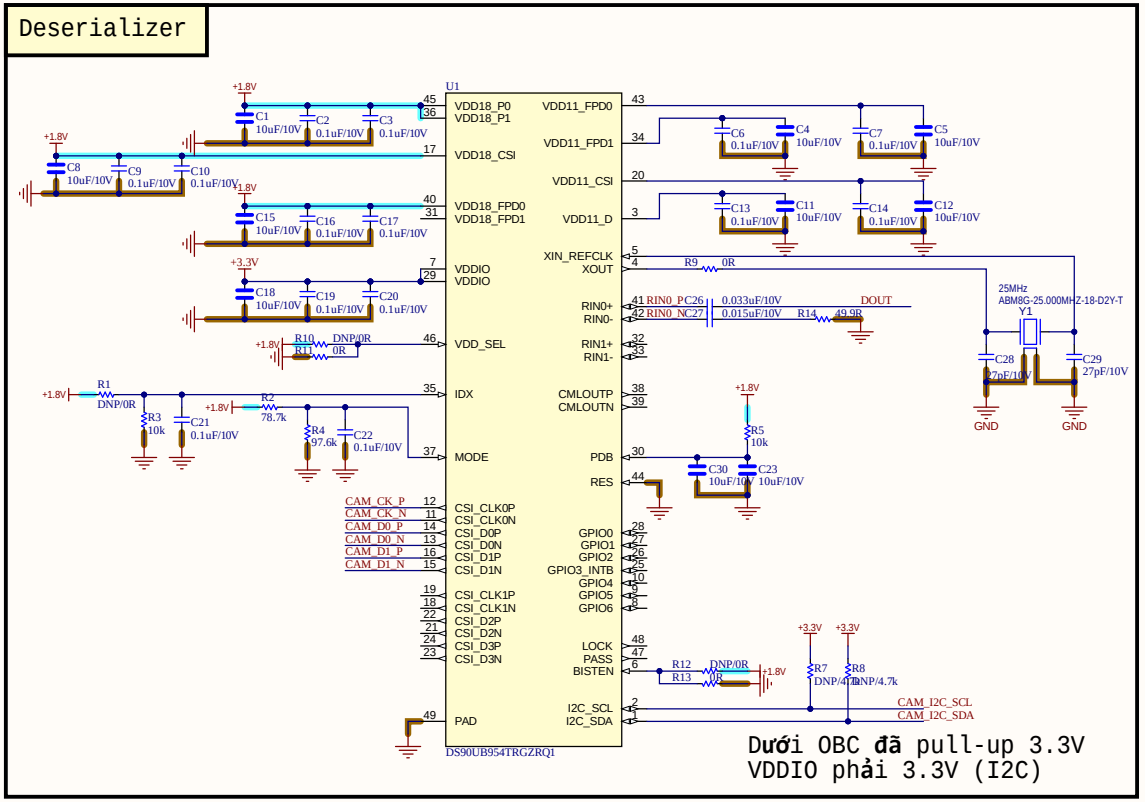
+3.3V POWER SUPPLY



PoC



Deserializer



Connector

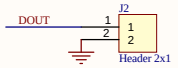


Table 7-2. Strap Configuration Mode Select

MODE NO.	V_{TARGET} VOLTAGE RANGE			V_{TARGET} STRAP VOLTAGE	SUGGESTED STRAP RESISTORS (1% TOL)		RX MODE
	V_{MIN}	V_{TYP}	V_{MAX}	VDD18 = 1.8 V	R_{HIGH} (kΩ)	R_{LOW} (kΩ)	
0	0	0	0.131 × $V_{(VDD18)}$	0	OPEN	10.0	CSI-2 non-synchronous Back Channel
1	0.179 × $V_{(VDD18)}$	0.213 × $V_{(VDD18)}$	0.247 × $V_{(VDD18)}$	0.374	88.7	23.2	RAW12 LF
	0.642 × $V_{(VDD18)}$	0.673 × $V_{(VDD18)}$	0.704 × $V_{(VDD18)}$	1.202	39.2	78.7	
2	0.296 × $V_{(VDD18)}$	0.330 × $V_{(VDD18)}$	0.362 × $V_{(VDD18)}$	0.582	75.0	35.7	RAW12 HF
	0.761 × $V_{(VDD18)}$	0.792 × $V_{(VDD18)}$	0.823 × $V_{(VDD18)}$	1.420	25.5	95.3	
3	0.412 × $V_{(VDD18)}$	0.443 × $V_{(VDD18)}$	0.474 × $V_{(VDD18)}$	0.792	71.5	56.2	RAW10
	0.876 × $V_{(VDD18)}$	$V_{(VDD18)}$	$V_{(VDD18)}$	1.8	10.0	OPEN	
4	0.525 × $V_{(VDD18)}$	0.559 × $V_{(VDD18)}$	0.592 × $V_{(VDD18)}$	0.995	78.7	97.6	CSI-2 Synchronous Back Channel

Table 7-15. Serial Control Bus Addresses for IDX

NO.	V_{IDX} VOLTAGE RANGE			V_{IDX} TARGET VOLTAGE	SUGGESTED STRAP RESISTORS (1% TOL)		PRIMARY ASSIGNED I2C ADDRESS	
	V_{MIN}	V_{TYP}	V_{MAX}	(V); VDD1P8 = 1.80V	R_{HIGH} (kΩ)	R_{LOW} (kΩ)	7-BIT	8-BIT
0	0	0	0.131 × $V_{(VDD18)}$	0	OPEN	10.0	0x30	0x60
1	0.179 × $V_{(VDD18)}$	0.213 × $V_{(VDD18)}$	0.247 × $V_{(VDD18)}$	0.374	88.7	23.2	0x32	0x64
2	0.296 × $V_{(VDD18)}$	0.330 × $V_{(VDD18)}$	0.362 × $V_{(VDD18)}$	0.582	75.0	35.7	0x34	0x68
3	0.412 × $V_{(VDD18)}$	0.443 × $V_{(VDD18)}$	0.474 × $V_{(VDD18)}$	0.792	71.5	56.2	0x36	0x6C
4	0.525 × $V_{(VDD18)}$	0.559 × $V_{(VDD18)}$	0.592 × $V_{(VDD18)}$	0.995	78.7	97.6	0x38	0x70
5	0.642 × $V_{(VDD18)}$	0.673 × $V_{(VDD18)}$	0.704 × $V_{(VDD18)}$	1.202	39.2	78.7	0x3A	0x74
6	0.761 × $V_{(VDD18)}$	0.792 × $V_{(VDD18)}$	0.823 × $V_{(VDD18)}$	1.420	25.5	95.3	0x3C	0x78
7	0.876 × $V_{(VDD18)}$	$V_{(VDD18)}$	$V_{(VDD18)}$	1.8	10.0	OPEN	0x3D	0x7A

SpaceCiinTech

Project:	Sub-system:	
Design by: *	Sheet title: Deserializer.SchDoc	Sheet 1 of 1
Checked by: *	Document No: 1	Size: A3
Approved by: *	Revision: *	Date: *

Name	Description	Designator	Footprint	LibRef	Quantity
10uF/10V	CAP High Value	C1, C4, C5, C8, C11, C12, C15, C18, C23, C24, C25, C30, C31, C32, C33, C34	CAP_C_1005X60	_CAP_C_High_2	16
0.1uF/10V	CAP	C2, C3, C6, C7, C9, C10, C13, C14, C16, C17, C19, C20, C21, C22, C35, C36	CAP_C_1005X60	_CAP_C_2	16
0.033uF/10V	CAP	C26	CAP_C_1005X60	_CAP_C_2	1
0.015uF/10V	CAP	C27	CAP_C_1005X60	_CAP_C_2	1
27pF/10V	CAP	C28, C29	CAP_C_1005X50_L	_CAP_C_2	2
1k@100MHz	FERRITE BEAD 600 OHM 0603 1LN	FB1	IND_C_1608X60	BLM18AG601SN1D	1
330R@100MHz	FERRITE BEAD 600 OHM 0603 1LN	FB2	IND_C_1608X60	BLM18AG601SN1D	1
47R@100MHz	FERRITE BEAD 600 OHM 0603 1LN	FB3	IND_C_1608X60	BLM18AG601SN1D	1
0532611171		J1	MOLEX_0532611171	Header 11x1_2M	1
Header 2x1		J2	JST_B2B-XH-A	Header 2x1	1
10uH		L1	IND730660x480	_IND	1
DNP/0R	RES	R1, R10, R12	RES_C_1005X40	_RES_B_2	3
78.7k	RES	R2	RES_C_1005X40	_RES_B_2	1
10k	RES	R3, R5, R6, R15	RES_C_1005X40	_RES_B_2	4
97.6k	RES	R4	RES_C_1005X40	_RES_B_2	1
DNP/4.7k	RES	R7, R8	RES_C_1005X40	_RES_B_2	2
0R	RES	R9, R11, R13, R16	RES_C_1005X40	_RES_B_2	4
49.9R	RES	R14	RES_C_1005X40	_RES_B_2	1
4.02k	RES	R17	RES_C_1005X40	_RES_B_2	1
DS90UB954TRGZRQ1	Dual 2 MP FPD-Link III deserializer with CSI-2 outputs for 2MP/60fps cameras & radar 48-VQFN -40 to 105	U1	QFN50P700X700X100-48N-EP410x410	IC_Deserializer_DS90UB954TRGZRQ1	1
TLV75518PDBVR		U2	SOT25	IC_POWER_AP7347DQ	1
TLV75733PDYDR		U3	SOT25-EPAD	IC_POWER_TLV75733PDYDR	1
Crystal 4 Pins	Crystal	Y1	CRYSTAL_ABM8G	OSC_24MHz	1